Project Proposal

Project Title: Intelligent Guitar Practice Tool

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Background and Motivation

There are plenty of Guitar learning tools in the market, each addressing different aspect of the idea of a practice tool. Platforms like Songsterr and Ultimate Guitar rely on community contributed tabs, which can vary significantly in accuracy and completeness. There are other examples, including one called Chordify, which generates chords of songs but doesn't provide tabs. AnthemScore provides AI-powered transcription to sheet music but struggles with guitar-specific notation and does not intelligently segment songs into practice sections. While each tool excels in particular areas, none combine automatic structural segmentation, accurate guitar tablature generation, and export to standard notation formats in a single, cohesive workflow. This fragmentation forces users to rely on multiple tools or manually create practice materials, creating a gap for an integrated solution that automates the complete project pipeline.

Table 1: Other Guitar Practice Tools features and weaknesses

<table>
<tr><td rowspan="2">Name</td><td>Features</td></tr>
<tr><td>Weaknesses</td></tr>
<tr><td rowspan="2">Songsterr</td><td>Generates AI tabs</td></tr>
<tr><td>Needs to be a YouTube link
Users mentioned poor accuracy</td></tr>
<tr><td rowspan="2">Chordify</td><td>Automatic chord detection from audio or YouTube videos</td></tr>
<tr><td>Provides only chord progression</td></tr>
<tr><td rowspan="2">Ultimate Guitar</td><td>Large Database of crowd sourced guitar tabs and chords</td></tr>
<tr><td>Has no option for automatic tab generation</td></tr>
<tr><td rowspan="2">AnthemScore</td><td>AI automatic transcription from audio to sheet
Exports to MusicXML, MIDI, and PDF formats</td></tr>
<tr><td>Poor at representing guitar techniques (bends, pull off, slides, hammer-ons)
No automatic segmentation for practice sessions</td></tr>
</table>

While some tools offer playback and practice features, none combine automatic segmentation with AI-powered transcription to generate practice-ready tablature that can be opened in standard music notation software.

This made me realize there is a gap in the market, a tool that could combine all the problems and create a solution which would allow both beginners and advanced players to be able to use a platform, which they can trust and learn their favourite songs from.

The tool will focus on the transcription and preparation phase of guitar practice. Once tablature is generated and exported to MusicXML, users can utilize existing music notation software (MuseScore, Guitar Pro) for playback, editing, and printing. This approach allows the project to focus on the core AI transcription challenge while using their preferred software for playback functionality.

Aims and Objectives

The main aim of the project is to develop an Intelligent Guitar Practice that automatically segments guitar recordings into practice friendly sections (riffs), generate AI-powered guitar tablature, and export the results in a format compatible with music notation software for playback and practice.

Major Objectives

1. Implement AI Guitar Transcription
   - 1.1. Integrate pitch detection model
   - 1.2. Provide support for both riffs and chords
2. Create Automatic Audio Segmentation
   - 2.1. Detect music phrases
   - 2.2. Identify repeating patterns
   - 2.3. Segment audio into sections
3. Generate Tab Output
   - 3.1. Link Guitar tab to the timing information
   - 3.2. Ensure compatibility with various guitar tab software (e.g. MuseScore, Guitar Pro)
   - 3.3. Export to MusicXML format
4. Build User Interface
   - 4.1. Allow users to upload audio files
   - 4.2. Display sections segmented
   - 4.3. Implement export to MusicXML
5. Test Systems Performance
   - 5.1. Test various guitar styles including Finger Style, Pick, Tapping
   - 5.2. Test on different skill level

    5.3. Conduct user testing with guitarists
    5.4. Measure transcription accuracy
    5.5. Compare with existing transcription tools
6. Document Findings and Process
    6.1. Produce Interim Report
    6.2. Produce Dissertation
    6.3. Produce Demonstration Video

Key Challenge

The primary technical challenge of this dissertation is implementing accurate AI guitar transcription. To address this, I have reviewed recent research in Automatic Music Transcription (AMT) [1], which has highlighted several key challenges specific to guitar audio: handling polyphonic music where multiple strings sound simultaneously, managing overlapping sound events with similar frequencies, and accurately detecting note onsets and offsets across different voices. Research in this field has explored various approaches including Neural Networks (NN), Non-negative Matrix Factorization (NMF), and more recent deep learning architectures [2]. Based on this analysis, I plan to utilise Spotify's Basic Pitch, which uses a deep Neural Network architecture combining CNNs for spectral analysis and Transformers for temporal modelling [3]. Basic Pitch would work perfectly for this project since it was trained on diverse music including guitar recordings from the GuitarSet dataset [4], showing strong performance on polyphonic guitar transcription.

Project Plan

The structure of the project plan is to start building a backend, following the order from the major objectives. The focus of the project is to build an MVP and ensure that we focus on the key components and add additional features once we have a complete project. We will work on the frontend UI/UX once that's done; this would give me time to sort out any issues that could occur.

The project plan has been designed to follow an agile model. With this, I would be able to take back the feedback from my supervisor and implement features seamlessly. I would also be able to listen to stakeholders, such as guitarists from different levels and take their suggestions into consideration when I produce my research.

Documentation

The 3 tasks that add up to 100% of the module's worth are the Interim Report (10%), Dissertation (75%), and Presentation (15%). I have given myself an ample amount of time for these documents based on their importance.

Research

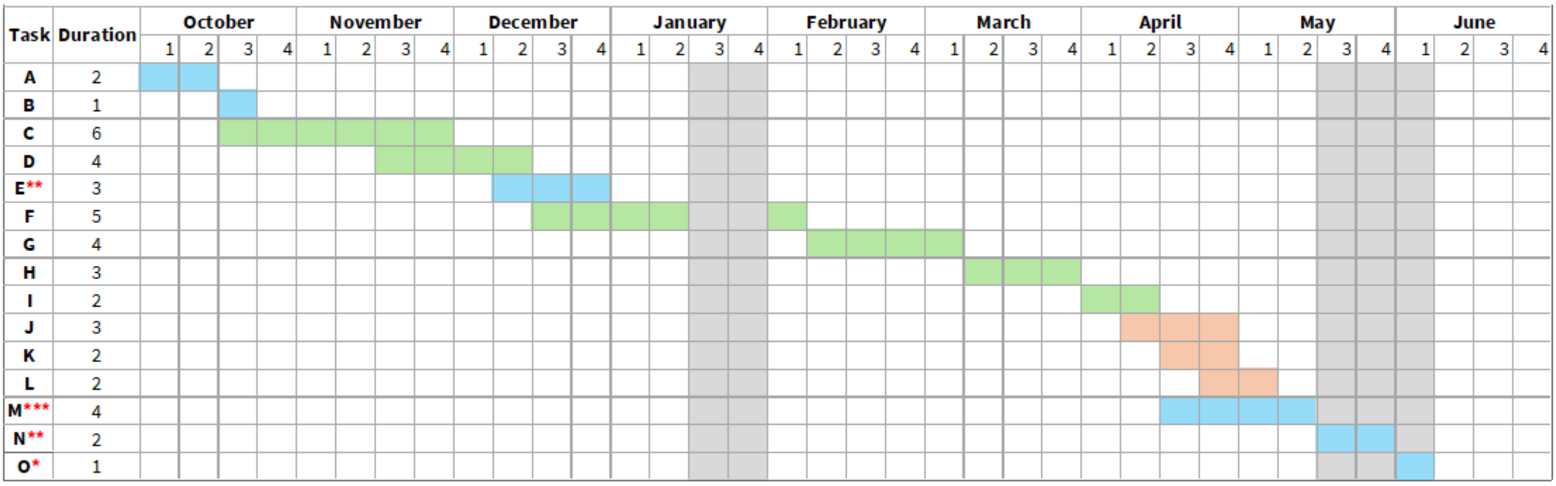
I will be conducting a survey using Microsoft Forms. This will be carried out before Christmas, so after January exams, I can begin making changes to the features and creating a UI based on the feedback provided. To formulate this research, I have completed the Ethics Form, the Data Management Plan, and will complete the ethics training.

Project Schedule

A Gantt Chart has been developed on the following page to showcase a general overview of the planned project schedule, allowing me to track the project status and to find whether the project is on schedule. As my module split has been split 70:50 with Autumn taking 70 credits, this has been accounted for during this project while also ensuring the project documents will be able to be submitted before the deadline. I have also considered other modules, exams and holidays that will take place and have allocated time to consider those situations, ensuring there is a contingency plan for unforeseen circumstances, including tasks taking longer than expected to carry out.

The project is also being tracked using the DevOps tool, Atlassian Jira. The project management tool will have a stronger focus on the individual tasks; therefore, keep a check on which task needs to be continued. Epics have been created, and problems have been broken down into tasks, which makes the tasks easier to complete and to feel a better accomplishment when they are done.

Table 2: Gantt chart for project management



ID	Task Description	Start	End	Duration
A	Complete Project Proposal	Oct W1	Oct W2	2 weeks
B	Complete Ethics and Data Management Plan	Oct W3	Oct W3	1 week
C	Foundation & Prototyping (Audio Loading, Basic Pitch)	Oct W3	Nov W4	6 weeks
D	Basic Tab Generation (Single Notes & Simple Melodies)	Nov W3	Dec W2	4 weeks
E	Write & Submit Interim Report (10% of grade)	Dec W2	Dec W4	3 weeks
F	Improve Tab Generation (Chords, Polyphonic Audio)	Dec W3	Feb W1	5 weeks
G	Automatic Audio Segmentation Implementation	Feb W2	Mar W1	4 weeks
H	Export Functionality for File Format	Mar W2	Mar W4	3 weeks
I	User Interface Development & Export to MusicXML	Apr W1	Apr W2	2 weeks
J	User Testing & Evaluation Study	Apr W2	Apr W4	3 weeks
K	Buffer Time for Exam Revision & Bug Fixes	Apr W3	Apr W4	2 weeks
L	Final Code Polish & Demo Video Creation	Apr W1	May W1	2 weeks
M	Write Final Dissertation (75% of grade)	Apr W3	May W2	4 weeks
N	Prepare Presentation & Demo for Assessment	May W3	May W4	2 weeks
O	Final Presentation/Demonstration (15% of grade)	Jun W1	Jun W1	1 week

Type	Colour
On-hold	
Documentation	
Development	
Testing	
Major deadlines	*

Bibliography

[1] E. Benetos, S. Dixon, Z. Duan, and S. Ewert, "Automatic music transcription: An overview," *IEEE Signal Processing Magazine*, vol. 36, no. 1, pp. 20-30, 2019.

[2] C. Hawthorne, E. Elsen, J. Song, A. Roberts, I. Simon, C. Raffel, J. Engel, S. Oore, and D. Eck, "Onsets and frames: Dual-objective piano transcription," in Proc. Int. Society for Music Information Retrieval Conf. (ISMIR), 2018, pp. 50-57.

[3] R. M. Bittner, J. J. Bosch, D. Rubinstein, G. Meseguer-Brocal, and S. Ewert, "A lightweight instrument-agnostic model for polyphonic note transcription and multipitch estimation," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 781-785.

[4] Q. Xi, R. M. Bittner, J. Pauwels, X. Ye, and J. P. Bello, "GuitarSet: A dataset for guitar transcription," in *Proc. International Society for Music Information Retrieval Conference (ISMIR)*, 2018, pp. 453-460.